EXPLORATORY MEETING · ELLEN WAN & ANDREW HOFFMAN

Clay, microbes and metal

How clay hydration and iron chemistry gate microbially influenced corrosion, and whether a small BSL-1 pilot is a shared question



Hunter Kinder · independent researcher · October 2026

START HERE

Three questions decide whether this is a fit for us

1

For Ellen

Is a clay–EPS–metal interface study a fit for your biofilm and EPS work, and with which BSL-1 strains?

2

For Andrew

Which corrosion-product signatures (FeS, Fe_3O_4, Cu_2S) would you trust from Raman, SEM/EDS and XRD after 2–4 weeks?

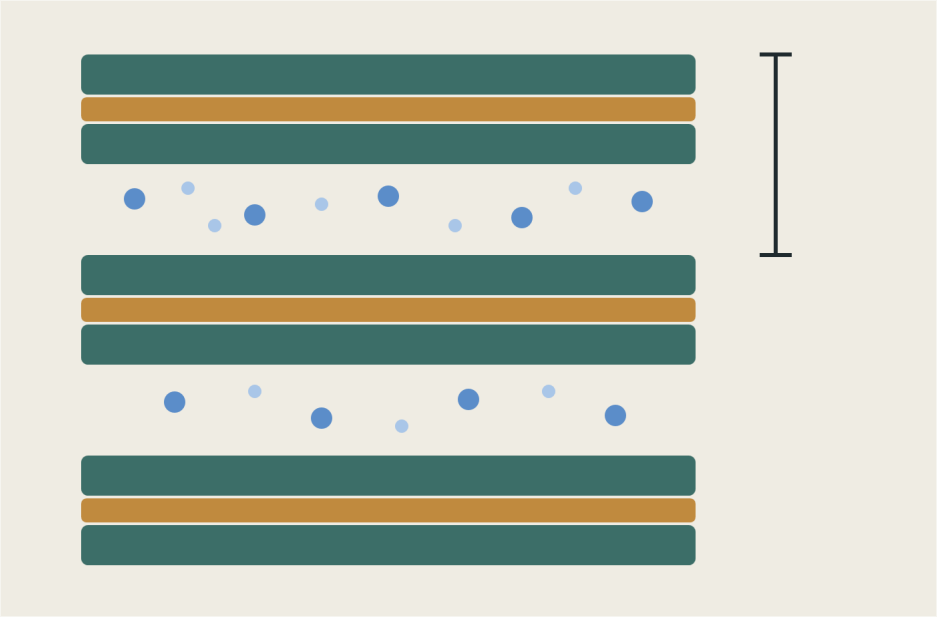
3

For both

What is the smallest result that would justify a 3–6 month study, and what would make you say no?

LITERATURE · MEASURED

Montmorillonite takes up water in discrete steps, and the cation sets which step



State	d001 spacing
Dry (0W)	≈10.0 Å
One water layer (1W)	≈12.6 Å
Two layers (2W)	≈15.2–15.3 Å
Three layers (3W, Na)	≈18–19.5 Å

At 40% RH, Na-montmorillonite sits near 12.5 Å (1W) and Ca-montmorillonite near 15.2 Å (2W). Only Na clays go on to osmotic swelling.

Ferrage et al. 2005a, Am. Mineral. 90:1358-1374; 2005b, Clays Clay Miner. 53:348-360. Schematic redrawn for this deck (green: silicate sheets; ochre: octahedral sheet; blue: cations and water). Bracket marks d001.

THE QUESTION I WOULD MOST LIKE YOUR VIEW ON

Screening, chelation and bridging are three different things, and only one changes connectivity

Mode	What the cation does	What would show it
Charge screening	Reduces repulsion between charged groups; links nothing	Surface potential shifts with ionic strength alone
Chelation	One ligand grips one ion at several points; still one chain	Coordination geometry, not network change
Bridging	One ion joins two separate chains, or a chain to a clay face	Rheology or retention changes beyond the ionic-strength effect

Denticity is not the same as connectivity: a monodentate contact can still be part of a bridge if the ion coordinates something else elsewhere. The clean contrast is a monovalent control against Ca^{2+} at matched ionic strength and pH.

Terminology after IUPAC Gold Book C01012 (chelation). A drop in surface potential alone would not establish cross-linking.

PROPOSED · NOT YET RUN

A 2–4 week pilot can show biotic versus abiotic, but not thresholds

Arm	Condition	What it answers
Slurry	MX-80 ≈1:10, live SRB, lactate and sulfate	Does sulfide form at all in this system?
Compacted	≈1.25 g/cm³ dry, live SRB	Does compaction change it at a reachable density?
Sterile clay	Autoclaved, same chemistry	Separates abiotic corrosion from microbial
No organism	Live clay, no inoculum	Tests the clay's own resident microbes

Readouts: weekly sulfide and pH; at the end SEM/EDS, Raman (mackinawite versus Cu_2S), XRD d001, and mass loss after standard cleaning.

Honest limit: at roughly 1 µm/yr, a few weeks of corrosion is below what a balance resolves. Sulfide and surface chemistry carry the result.

Design after SKB TR-16-09 and Martinez-Moreno et al. 2023, minus the ^{35}S radiotracer. Triplicate arms. All BSL-1, non-radioactive.

PROPOSED MEASUREMENT LAYER · FLOW CYTOMETRY

Flow cytometry adds cell-state observations to the interface model

It measures suspended-cell phenotypes; by itself, it does not preserve where cells sat inside clay or a biofilm.

WHAT IT CAN MEASURE

- Gated cell-event counts and concentration using counting beads
- Dye-defined membrane-integrity classes, after strain-specific validation
- Scatter and per-cell fluorescence distributions
- Optional metabolic / ROS panels only after assay validation

WHAT IT CONSTRAINS

- Suspended and detached-cell abundance
- Fractions in measured physiological-state gates
- Net growth/death or attachment/detachment rates when time series are available
- Equifinality: similar bulk chemistry can arise from different cell-state histories

LIMITS

- Spatial position in the clay pore network
- EPS cross-link density or mineral phase
- Corrosion rate / causal mechanism
- “Live” status from fluorescence alone without controls; clay particles and aggregates can distort gates

MINIMUM VIABLE ADD-ON: absolute counts + a validated membrane-integrity panel at preselected timepoints; keep maximum pit depth as the primary corrosion endpoint.

Sources: Kerstens et al. (2015), doi:10.1007/s12223-015-0400-4; Grainha et al. (2020), doi:10.3390/antibiotics9110741; Yip et al. (2022), doi:10.1371/journal.pcbi.1010533.

SCOPE AND SEPARATION

The work stays BSL-1, non-radioactive and fully separate from my employer

Biosafety

- DSMZ lists the candidate sulfate reducer (DSM 644, = ATCC 29579) as Risk Group 1
- One supplier page carries a conflicting product label, so I will confirm with them and your IBC before ordering
- Note the genus was revised to *Nitratidesulfovibrio* in 2020
- IBC approval before any organism is handled

Separation

- No radioactivity anywhere in the design
- No employer facilities, data, imagery or branding
- Outside-activity disclosure in progress with my supervisor
- Everything here is public literature or my own labelled models

THE ASK

I bring modelling and analysis, and I would need bench access and mentorship

What I bring

- Reactive-transport and surface-complexation modelling
- Julia numerics and data analysis
- Analytical chemistry background: HPLC, ICP, GMP and GLP lab work
- A pre-registered readout plan, with every claim typed as measured, modelled or assumed

What I would need

- Anaerobic chamber and BSL-1 bench time
- Access to SEM/EDS, XRD and Raman
- Mentorship on biofilm and corrosion method choices
- Honest feedback on whether this is worth pursuing at all

Self-funded, evenings and weekends. Consumables for the pilot are plausibly a few thousand dollars, which is my estimate rather than a quote.